Machine learning is a subset of artificial intelligence that focuses on algorithms learning patterns from data. Modern systems rely on large corpora and gradient based optimization methods.

Natural language processing studies the interaction between computers and human language. It spans syntax, semantics, and pragmatic inference across many application domains.

Computer vision enables machines to interpret visual scenes from digital images. Convolutional networks drove remarkable progress on recognition benchmarks during the last decade.

Reinforcement learning trains agents to make sequences of decisions by maximizing cumulative reward signals. Applications include game playing, robotics, and recommendation systems.

Transfer learning reuses knowledge from a source task to improve learning on a related target task. Pretrained language models are a prominent example of this paradigm in practice.